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24th April, 2026

Health Information Systems

Decision Support System

CxRisk: Cervical Cancer Risk Stratification and Screening Triage

Introduction

Cervical cancer, a slow growing cancer, is the fourth most common cancer affecting women and representing a significant health and mental burden in the United States. According to the National Cancer Institute, approximately 13,490 new cases of invasive cervical cancer will be diagnosed in the United States in 2026 (NCI SEER, 2025). A Pap test, a procedure in which cells are scraped from the cervix and observed under a microscope, can detect the cancer. The SEER database reports an estimate of 0.6% of women that will be diagnosed with cervical cancer in their lifetime (NCI SEER).

Cervical cancer is often attributed to a long precancerous phase, meaning it can be detected and treated effectively if detected. The issue lies with cervical cancer being asymptomatic in early stages. However, with proactive measures in place, cervical cancer can be treated when caught and cared for early on. The primary cause of cervical cancer is persistent infection with high-risk strains of human papillomavirus (HPV), notably HPV16 and HPV18 (Zhang et al., 2020). Cofactors accelerating the progression of cervical cancer include smoking, long-term use of hormonal contraceptives, increased sexual history, a weakened immune system, and many more (American Cancer Society, 2025).

While most cases of asymptomatic HPV are cleared naturally by the immune system, persistent infection of certain types of HPV may develop into cancer. In normal conditions, it

may take 15-20 years for abnormal cells in a woman's body to turn into cancerous cells. However, if she has untreated HPV, which results in a weaker immune system, she may see cancerous cells develop in 5-10 years (WHO, 2025).

To reduce the likelihood of developing cervical cancer, there are many preventative measures in place to help prevent cervical cancer, including improving public awareness, health literacy, and access to information and services. Coupled with the HPV vaccination for young girls ages 9-14, regular cervical screenings every 3-5 years are essential to detecting cervical cancer (WHO, 2025). Once detected, cases in the early stage are more likely to be treated and overcome (NIH, 2025).

While HPV/Pap testing, routine tests, and colposcopy procedures have reduced rates of cervical cancer mortality, there are many structural screening challenges including missed appointments for routine testing and predictive tools relying on testing rather than using behavioral risks to flag patients (ASCCP, 2019). This introduces the prospect of using Health Information Technology (HIT) solutions embedded in Electronic Health Records (EHRs) to conduct initial screening and recommend next steps in treatment. By integrating machine learning and predictive analytics with behavioral risk factors and a clinician-centered workflow in mind, the HIT Decision Support System (DSS) CxRisk can help thousands of women discover and treat cervical cancer in its early stages with more targeted screening interventions.

Stakeholders

In addition to affecting patient outcomes, cervical cancer influences the roles of healthcare systems, clinicians (OB-GYNs, PCPs, NPs), nurses, medical assistants, clinic managers and quality staff, health insurance payers, and IT and compliance teams. Digital

screening solutions are growing not only in the market but in popularity in their effectiveness and alleviating clinician burnout and administrative headaches. CxRisk involves all of these stakeholders to reduce the strain caused on one department:

1. Clinicians (OB-GYNs, PCPs, NPs) are the primary users of CxRisk, interpreting the model determined risk score that fits into the existing workflow. Keeping clinicians in control, they have the ability to override the suggested triage recommendation.
2. Nurses and medical assistants are responsible for inputting or updating structured patient data into CxRisk with quick lookup capabilities.
3. Clinic managers and quality staff can use the population dashboard to monitor referral volumes, tool performance, population demographics, etc.
4. Health insurance payers are less likely to reject claims with ICD-10 codes directly built into the system.
5. The IT and compliance departments are responsible for ensuring CxRisk meets HIPAA, interoperability, and cybersecurity requirements.

Through decision support IT solutions, healthcare providers can deliver personalized treatment plans, improving patients outcomes and the mortality rate of women with cervical cancer.

Policy and Management Issues

In the healthcare space, it is essential for CxRisk to follow clinical guidelines to ensure safe, effective, and scalable outcomes for patients. Most notably is compliance with HIPAA policies to protect patient data leakage, protected with strong encryption and controlled access. Additionally, CxRisk must follow screening guidelines outlined by the ASCCP Risk-Based

Management Consensus Guidelines most recently updated in 2019. To assist in reducing alert fatigue, the DSS solution is designed to have few buttons, a clean design, and easily identifiable results to ensure a smooth care delivery process. And lastly, the CxRisk integrates LOINC, SNOMED, and ICD-10 codes to seamlessly integrate with lab results and payers.

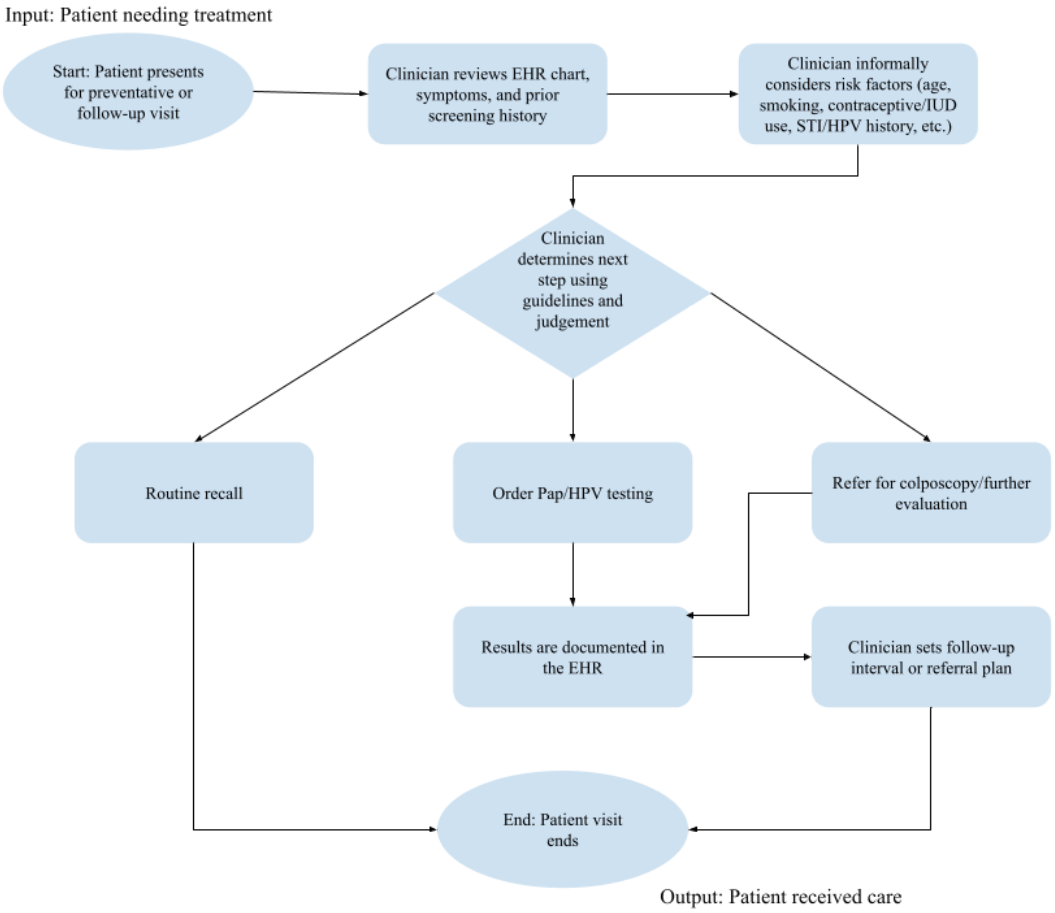
Existing Software Vendor and Product

To assist clinicians in detecting cervical cancer, the ASCCP has developed the ASCCP Management Guidelines App using both current and past test results to create personalized risk assessments for a patient. The app streamlines data from the ASCCP Risk-Based Management Consensus Guidelines, determining a patient's immediate risk of precancer (CIN3+) or 5-year risk of progressing to cancer (BD, 2022).

While the app follows clinical guidelines, it's limited by its requirement of a prior Pap or HPV test result to determine risk assessments, using rule-based logic and failing to account for behavioral variables that may trigger high-risk patients. The DSS model was trained to address this gap by incorporating behavioral variables like smoking, STI history, and contraceptive use to flag high-risk patients even before they conduct any tests.

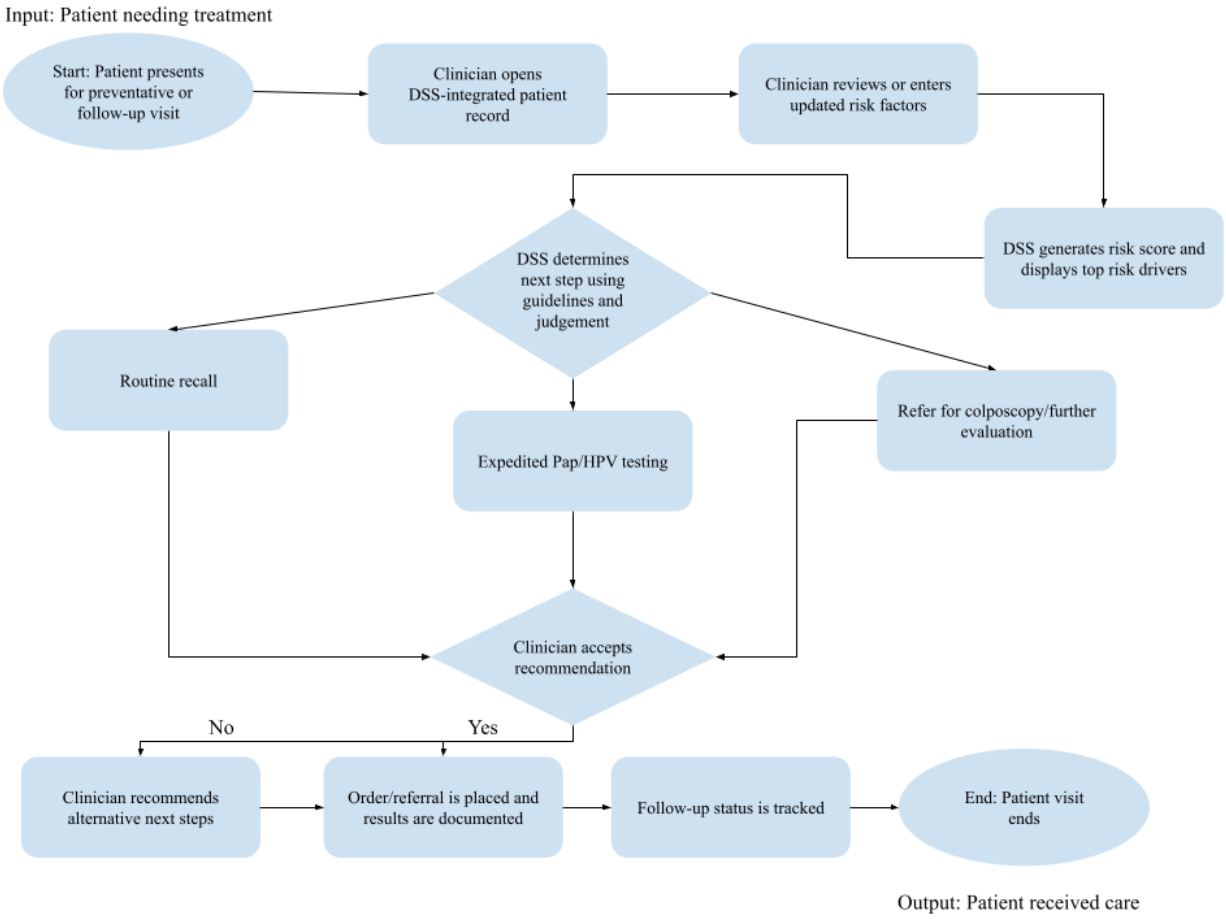
A second gap we plan to resolve is that the 2019 ASCCP guidelines cannot cover all clinical situations by ensuring proper training of the model to cover multiple combinations of risk factors while simultaneously protecting the results from deviating due to missing data (Perkins et al., 2020).

Status Quo Process Model



At present, cervical cancer screening and follow-up are largely guideline-based and clinician-driven. During a visit, the clinician reviews age, prior Pap/HPV history, symptoms, and any known risk factors, then decides whether the patient should remain on routine screening, receive further HPV/Pap testing, or be referred for colposcopy. In practice, this status quo process depends heavily on manual chart review and provider judgment rather than a standardized risk score, which can make triage less consistent when visit time, documentation quality, or clinic capacity are limited.

DSS-Enabled Process Model



The DSS changes this workflow by standardizing the triage step with a data-driven recommendation. Instead of relying only on informal judgment, the clinician reviews or enters structured risk-factor information from the chart and visit, such as age, smoking history, contraceptive/IUD use, STI history, and prior diagnosis indicators, and the system calculates a predicted probability of an abnormal screening or diagnostic outcome. The DSS then displays the main contributors to risk and recommends the most appropriate next action while still allowing the clinician to override the recommendation and document the reason.

DSS Components and Architecture

The architecture of CxRisk consists of three core components: (1) a patient risk-factor database, (2) a predictive modeling engine, and (3) a clinician-facing recommendation interface. Clinicians log in securely with their IT-assigned username and password through an R Shiny interface. Upon initial patient arrival, nurses can input new patient data or extract existing records, automatically initiating data retrieval and updating processes. At the physician assessment stage, the physician runs the assessment model, a Random Forest model, which processes the patient's data and outputs a risk score with an associated level for ease of interpretation. Along with the risk score are the patient's top drivers of risk to direct the clinician's attention. The clinician is lastly presented with a triage recommendation of the patient's next steps with the ability to override the recommendation from experienced, clinical judgement.

To include HIPAA standards of data privacy and protection, patient data is only revealed to users with proper clinical clearance. Administrators (managers, staff, etc.) will only have access to the population dashboard upon logging into CxRisk. Clinicians (physicians, nurses, etc.) will only have access to the patient search and risk assessment pages, having no administrative requirement to view population data.

I. Explain the major components of your IT solution and their content by mapping the decision support requirements of the problem, including the impact of standards and HIPAA, on the technology architecture of your IT solution

Data Sources and Analysis

The main project data source is the Cervical Cancer Dataset sourced from theUniversity of California, Irvine (UCI) Machine Learning Repository, which corresponds to the public cervical cancer risk-factor dataset used to train and test the prototype DSS (Fernandes et al., 2017). The dataset consists of 858 patient instances each with 32 features and 4 binary classification labels to indicate presence of cancer depending on different tests conducted. The behavioral risks in the dataset include smoking, pregnancies, number of sexual partners, use of hormone contraceptives, etc., along with clinical information regarding specific strains of STDs to obtain a clear picture of the patient's history.

1. Data Pre-Processing
    - 1.1. Two features with >50% missingness were dropped, while all other missing values were imputed with the median for continuous variables and the mode for binary variables.
    - 1.2. We have engineered two composite features: (1) STD_burden (the sum of all twelve individual STD indicator columns to create a composite measure avoiding multicollinearity) and (2) AgeBucket (binned age categories for stratified comparisons).
    - 1.3. As part of cluster analysis, RiskCluster demonstrated strong correlation among patients in High, Medium, or Low risk levels, creating meaningful grouping.
    - 1.4. Ultimately, the final set of features used in model training and evaluation is: Age, AgeBucket, Number of Sexual Partners, Number of Pregnancies, Smokes (years), Hormonal Contraceptives (years), IUD (years), STD_burden, Dx:Cancer, Dx:HPV, and RiskCluster. These factors are not only indicators of previous sexual

infections (Number of sexual partners, STD_burden) but also of likelihood of diagnosis (Age, AgeBucket) and behavioral contributions (Smokes (years), Number of Pregnancies).

2. Data Analysis

2.1. Through univariate testing, we have discovered that the population of women with cervical cancer has a median of 25 years, with 75% of patients under the age of 32 and creating a right-skewed distribution with a long tail extending to age 84.

2.2. Coinciding with real-world clinical incidence (0.6% of women experiencing cervical cancer), all four screening outcomes are rare, motivating SMOTE oversampling and class-weight adjustments conducted during model development.

Outcome	Positive Rate	Clinical Meaning
Hinselmann	4.1%	Abnormal colposcopy
Schiller	8.6%	Abnormal iodine staining
Citology	5.1%	Abnormal Pap smear
Biopsy	6.4%	Confirmed histological abnormality
AnyAbnormal (composite)	11.9%	At least one positive test

2.3. In our bivariate analysis between risk factors and outcomes, we discovered systematic differences across all key continuous features when comparing Biopsy = 0 and Biopsy = 1. Namely, the STD_burden shows the most pronounced difference. Patients with an abnormal biopsy have 5.5x the STD burden of patients with a normal biopsy, signaling the strongest bivariate signal in the dataset.

Feature	Biopsy=0 Mean	Biopsy=1 Mean	Direction
Age	26.6 yrs	31.4 yrs	Higher in abnormal
Number of sexual partners	2.4	3.1	Higher in abnormal
Num of pregnancies	2.1	3.2	Higher in abnormal
Smokes (years)	0.8	2.1	Higher in abnormal
Hormonal Contraceptives (years)	2.3	3.8	Higher in abnormal
STD_burden	0.15	0.82	Much higher in abnormal

3. Machine Learning Model Development

   3.1. Logistic Regression: With a balanced class weight, this model was an interpretable baseline providing odds ratios for direct clinical meaning.

   3.2. Random Forest + SMOTE: This model captures non-linear interactions, with SMOTE synthetically oversampling the minority class to 50% in the training set.

   3.3. Comparison: The Random Forest model achieved a higher AUC (0.632 vs. 0.584) and is selected in the DSS as the operational model for triage scoring. In a traditional clinical setting, recall is prioritized over precision to avoid missing a true abnormal case, so the threshold was lowered to 0.3 (from a default of 0.5) to improve sensitivity. Through odd ratio (logistic regression) and permutation feature importance (random forest) analysis, we found that both models highlighted STD_burden as the dominant clinical predictor.

4. Triage Assignments using the Random Forest + SMOTE Model

   4.1. Patients are categorized into low, medium, and high risk groups with clear probability cutoffs, highlighting the proper triage recommendations for each group. High-risk patients should be referred for a colposcopy, medium-risk patients for expedited testing to clarify risk, and low-risk patients are recommended to follow up with routine screening.

Category	Threshold	Action	Clinical Rationale
Routine Screening	$p < 0.15$	Standard annual/biannual follow-up	Low-risk; base rate below population average
Expedited Testing	$0.15 \leq p < 0.40$	Priority appointment within 4 weeks	Elevated risk; warrants closer monitoring
Referral	$p \geq 0.40$	Immediate specialist referral	High-risk; clinical evaluation required

Solutions for "What-if" Scenarios

To evaluate the sensitivity and robustness of the model for CxRisk, we conducted thorough "What-if" analysis.

1. Input Uncertainty
    - 1.1. To test input uncertainty, smoking status was intentionally misreported (non-smoking patients were assigned a smoking duration), which led to a change in probability of nearly 0 due to some patient's risk level rising while others fell.
    - 1.2. STD history misreporting led to a 70.7% of patients switching triage category (changing STD_burden from 0 to 2), confirming that STD history is the most clinically critical data element and its accuracy determines the integrity of the triage system.
        - 1.2.1. Clinical Implication: Clinics should invest in systematic, private, and structured STD collection protocols.
2. Missing Data Robustness
    - 2.1. AUC Degradation Under Increasing MCAR Missingness
        - 2.1.1. The model degrades 0.059 AUC units from 0% - 30% random missingness, which is a relatively small and smooth decline. However, the standard deviation increases significantly from 0.000 to 0.079 as

missingness increases, meaning that prediction consistency for individual patients degrades significantly even if the aggregate AUC holds.

2.1.2. Clinical implications: A patient with 30% of their record missing may receive different triage assignments across repeated evaluations.

2.2. Per-Feature AUC Drop When Fully Masked

Missingness Level	Mean AUC (20 repeats)	Std Dev	AUC Drop from Baseline
0% (baseline)	0.632	0.000	—
5%	0.627	0.023	-0.005
10%	0.586	0.044	-0.046
15%	0.610	0.039	-0.022
20%	0.585	0.065	-0.047
25%	0.593	0.059	-0.039
30%	0.573	0.079	-0.059

3. Threshold Optimization

3.1. The cost function $C(T) = \alpha(FN(T)) + \beta(FP(T))$ was used to evaluate across thresholds $T \in [0.05, 0.80]$ under three alpha values (relative cost of missed case) and three capacity scenarios (maximum referral rate the clinic can absorb).

3.2. The key findings indicate that under tight capacity (<5% referrals), the optimal threshold rises to 0.71 across all alpha values, inferring that the clinic is forced to accept substantially more missed cases to stay within referral capacity. The recommendation is to use T=40 as the default referral threshold for routine operations.

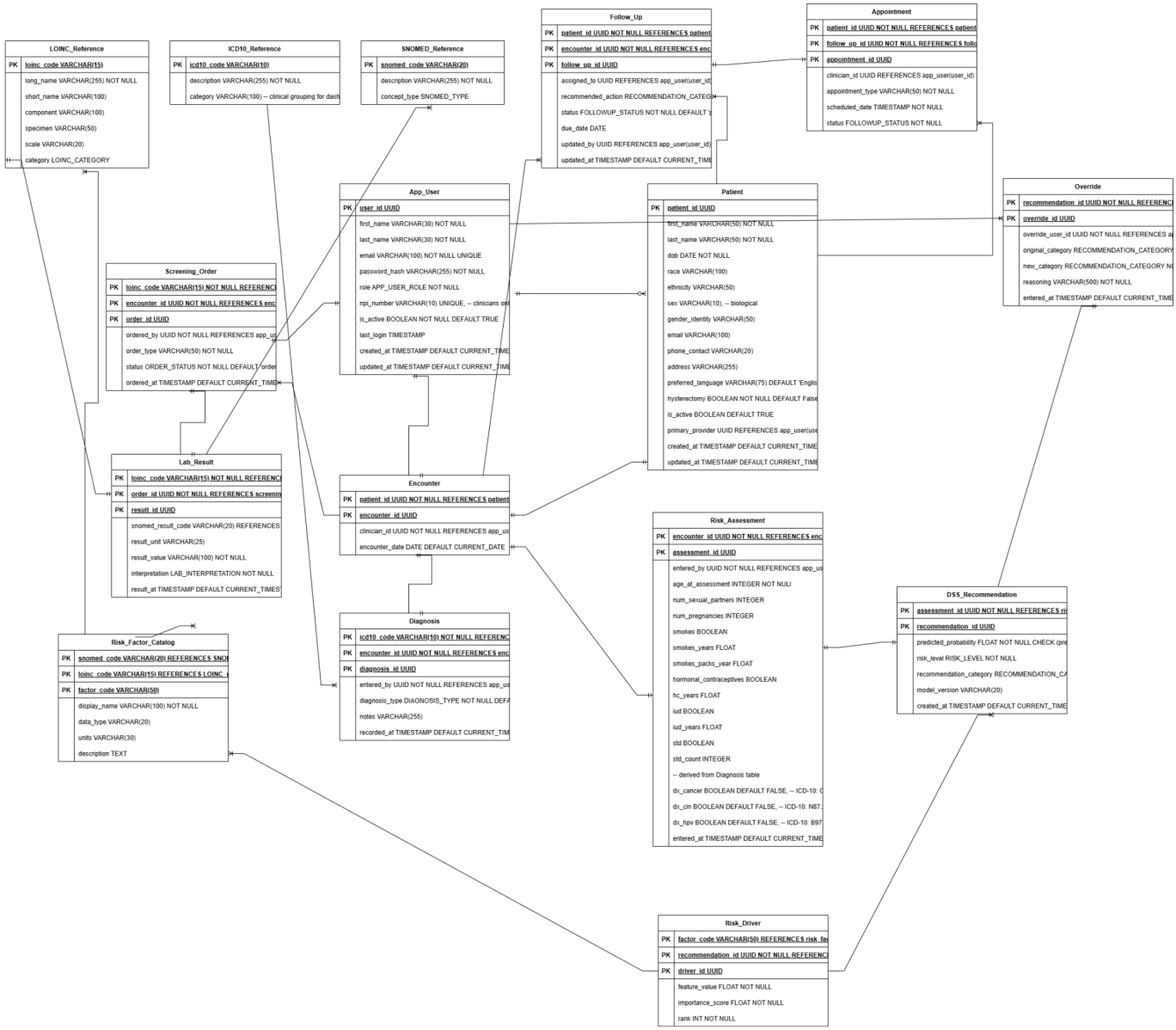
4. Summary of All What-if Findings

Scenario	Key Result	Operational Recommendation
A1 — Smoking flip	11% triage switches; modest mean $\Delta p = -0.007$	Smoking data: low priority for data quality investment
A2 — STD flip	70.7% triage switches; mean $\Delta p = +0.223$	STD data: highest priority — mandatory intake field
B — Missingness 0→30%	AUC drops 0.059; SD doubles	Flag 'Number of sexual partners' and 'STD_burden' as required fields
B — Per-feature masking	Masking 'partners' alone = -0.088 AUC	Two fields alone account for >15% of model's discriminative power
C — Threshold sweep	Optimal $\tau = 0.40$ (moderate), 0.71 (tight capacity)	Use τ=0.40 as default; adjust to 0.71 under staffing constraints
Tornado	STD_burden drives 47% probability range	Data quality of STD history determines triage integrity

Database Structure

The Patient entity stores demographic details and links to exactly one Physician, forming a one-to-many relationship. Each Encounter is tracked between one patient and one clinician, which outputs an individualized Risk Assessment, generating a DSS Recommendation, and pulling out key Risk Drivers. An app user can be a clinician, nurse, or administrator. Clinicians can order screening tests and see test results that correlate with LOINC_Reference, ICD10_Reference, and SNOMED_Reference.

While the nature of the database is complex, this is abstracted in the user-interface to provide a seamless patient-provider interaction.



Implementation of PEIT

Using PEIT here as People, Environment, Information, and Technology, the framework fits this project well because the DSS is not just a model; it is a clinical workflow tool that depends on who uses it, where it is used, what information it consumes, and how the technical system delivers recommendations. That matches the project proposal, which defines the system as a combination of a patient risk-factor database, an interpretable predictive model, and a

workflow-oriented interface that recommends routine screening, expedited HPV/Pap testing, or referral for further evaluation under capacity constraints. The revised workflow from Homework 6 also shows that the DSS sits directly in the clinician decision process, with the clinician retaining the ability to confirm or override the recommendation.

1. People:
    1.1. The primary users are clinicians and care teams, with secondary users such as clinic managers and quality staff. Implementation should begin with stakeholder mapping and role definition: clinicians use the risk score and recommendation at the point of care, nurses or medical assistants verify structured inputs, and managers monitor referral volume, abnormal-result follow-up, and capacity impacts.
    1.2. This component is appropriate because the proposal emphasizes transparent risk drivers and clinician-facing recommendations rather than full automation, so adoption depends on trust, training, and clear accountability. In practice, deployment should start with training sessions, short workflow simulations, and explicit guidance that the DSS is advisory, not a replacement for clinical judgment. That aligns with health IT implementation guidance emphasizing project management, communication, training, and user adoption as core parts of CDS rollout.
2. Environment:
    2.1. The environment for this DSS is a resource-constrained outpatient screening setting where appointment availability, diagnostic capacity, and patient outreach

limitations matter. For that reason, the first deployment should be a pilot in one clinic or one screening workflow rather than a full enterprise rollout.

2.2. The pilot should define where the tool appears in the visit, what happens when data is missing, and how referrals are queued when capacity is limited. This component is appropriate because your proposal explicitly frames triage decisions as capacity-sensitive and because the existing process model in Homework 6 shows the DSS embedded between chart review and follow-up action. A staged pilot-to-rollout approach is also consistent with ONC/CDC CDS integration guidance.

3. Information:

3.1. The information layer includes the cleaned dataset, structured risk-factor fields, outcome labels, missingness indicators, risk thresholds, and explainability outputs. The current project already separates data extraction/cleaning, database setup, model building, and front-end development, which makes it sensible to deploy information assets in phases: first the cleaned nullable and analysis-ready datasets, then validated model outputs, then clinician-facing explanations.

3.2. This component is especially appropriate because the proposal says the model must explicitly address missingness and self-report uncertainty, and the cleaned files you created already operationalize that with derived fields, missingness handling, and an analysis-ready version. Before go-live, the team should finalize a data dictionary, lock the production feature list, define threshold rules for routine screening versus expedited testing versus referral, and create a monitoring plan for calibration drift and override patterns.

4. Technology:

    4.1. The technology layer includes the database, model service, front-end, access controls, audit logs, and integration with the clinic workflow or EHR-like interface. For this project, the best deployment path is a modular architecture: database tables for patient factors and results, a model layer that can run logistic regression and gradient boosting, and a UI that shows risk probability, key drivers, missing-data flags, and the recommended next step.

    4.2. This is appropriate because the proposal explicitly compares multiple models and prioritizes interpretability and robustness, while the Homework 6 process model requires the recommendation to be visible at the exact decision point where the clinician can accept or override it. Technically, the rollout should proceed in four steps: sandbox testing, pilot with retrospective cases, limited live pilot with clinician override required, and then broader deployment with ongoing monitoring.

A concise deployment plan would therefore be: Phase 1: preparation (finalize cleaned data, thresholds, interface mockups, training materials); Phase 2: validation (retrospective testing on historical records and edge cases); Phase 3: pilot (single clinic, clinician override mandatory, track referrals and missed abnormal cases); and Phase 4: scaled deployment (expand only if usability, safety, and capacity metrics remain acceptable). This PEIT structure is appropriate because it treats implementation as a socio-technical change effort rather than just a modeling exercise.

Policy Implications

The DSS process raises important health IT policy issues because cervical screening records are sensitive health information. HIPAA compliance, patient consent, encryption, access controls, and audit logging are essential. Policies also need to clarify how DSS recommendations should be used in clinical workflow. The system should be treated as advisory, with clinicians retaining final authority. From a public health perspective, risk-stratified triage could strengthen cervical cancer prevention by helping clinics identify patients who need faster screening or follow-up.

Additional Benefits and Challenges of Gen AI and LLMs

GenAI and LLMs could add value to this DSS in several specific ways without replacing the core predictive model. First, they could generate a clinician-facing explanation of the recommendation in plain language, for example: “This patient was flagged for expedited testing because age, smoking history, STI-related variables, and prior abnormal indicators jointly increased predicted risk.” Second, they could produce patient-friendly outreach text or after-visit instructions tailored to literacy level and preferred language, which would be useful when the recommended next step is HPV/Pap testing or referral. Third, they could help with documentation, such as drafting a note that summarizes the model output, the main risk drivers, whether data were incomplete, and why the clinician accepted or overrode the recommendation. Fourth, they could support a guideline Q&A assistant layered on top of approved screening protocols and clinic policies so staff can ask operational questions without manually searching documents. These use cases fit your project because the proposal already emphasizes transparent risk drivers, mitigation strategies, and population-level insights rather than just a raw score.

The main challenges are important. LLMs can hallucinate, oversimplify clinical nuance, or present uncertain statements too confidently. They can also drift away from approved screening guidance if they are not constrained to trusted sources. In addition, they introduce privacy and security concerns because any prompt containing patient data becomes part of the health IT workflow and must be handled appropriately. There is also a fairness risk: a patient-facing LLM may generate less clear or less culturally appropriate language for some groups unless it is tested carefully. For those reasons, GenAI in this project should be limited to explanation, communication, and documentation support, while the actual triage logic remains rule-based and model-based. A safer design would use retrieval from approved guideline content, human review for outward-facing messages, audit logging, and a strict ban on letting the LLM independently determine screening or referral status.

Conclusions

CxRisk demonstrates that a behavioral risk-factor-driven DSS can meaningfully extend the reach of existing cervical cancer screening tools by identifying high-risk patients before any clinical test result is available. The ASCCP Management Guidelines App, while evidence-based and clinically valuable, is fundamentally reactive as it requires prior Pap or HPV results to function. CxRisk addresses this gap directly by training a Random Forest model on behavioral and demographic variables including STD burden, smoking history, contraceptive use, and sexual history, enabling pre-test triage in populations where test-based stratification fails most acutely.

The what-if analysis confirmed both the model's robustness and its vulnerabilities. STD burden emerged as the single most clinically critical data element, with misreporting causing

70.7% of patients to switch triage categories, which underscores that the system's integrity depends as much on data collection protocols as it does on model performance. The model degrades gracefully under random missingness up to 30%, but prediction consistency for individual patients declines significantly, reinforcing the importance of complete intake documentation.

The PEIT implementation framework and four-phase deployment plan ensure that CxRisk is treated as a socio-technical change effort rather than a purely technical deployment. Clinicians retain full override authority, role-based access controls enforce data privacy, and the population dashboard gives managers the visibility needed to monitor both tool performance and equity outcomes over time.

Specific recommendations

1. **Prioritize structured STD history collection at intake.** Because STD burden is the dominant predictor and its misreporting destabilizes triage assignments, clinics should implement standardized, private intake protocols for sexual health history before deploying CxRisk.
2. **Use T = 0.30 as the default operational threshold for routine settings**, lowering it to maximize sensitivity and reduce missed high-risk cases. Reserve threshold adjustment to 0.71 only in documented capacity-constrained scenarios where referral volume must be capped below 5%.
3. **Mandate clinician override documentation with a reason code from day one of the pilot.** Override data is essential for ongoing model improvement, calibration monitoring, and institutional liability protection.

4. **Deploy first in a single safety-net or FQHC clinic** where the gap between behavioral and test-based risk stratification is largest and the patient population stands to benefit most from pre-test flagging.
5. **Establish a formal model monitoring protocol before go-live**, with quarterly calibration checks and a defined AUC degradation threshold (e.g., drop > 0.05 from baseline) that triggers recalibration or model retraining.
6. **Re-collect behavioral variables at every visit**, not only at initial intake. Since risk profiles change with new partners, smoking cessation, or new STI diagnoses, a static intake record will produce inaccurate recommendations for returning patients over time.
7. **Restrict any LLM integration strictly to explanation, communication, and documentation.** Triage logic must remain model-based. All LLM outputs touching patient data must route through HIPAA-compliant, BAA-governed APIs with human review for patient-facing content.
8. **Conduct an equity evaluation at 12 months post-deployment**, comparing DSS recommendation rates, clinician acceptance rates, and clinical outcomes across racial, ethnic, and insurance subgroups to identify and correct any disparities the tool may introduce or reinforce.

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